

Noah Oatley 101189707 Assignment 5

Step 1: Memcheck

```
==47674==  
==47674== HEAP SUMMARY:  
==47674==    in use at exit: 0 bytes in 0 blocks  
==47674==   total heap usage: 2,011 allocs, 2,011 frees, 156,384 bytes allocated  
==47674==  
==47674== All heap blocks were freed -- no leaks are possible  
==47674==  
==47674== For lists of detected and suppressed errors, rerun with: -s  
==47674== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)  
noatley@Ubuntu:~/3303/Assignment1$
```

Above is the output of calling the memcheck. No memory errors are present, so no changes had to be made.

Leaking memory is bad for embedded systems as they have limited RAM due to size constraints. They are typically hard-time constrained, and memory issues can cause real-time responses to be delayed. They are also enabled/running for a long time, which if a memory leak is present, can cause the device to run with a bad error for a long duration.

Step 2: DRD

```
==48402==  
==48402== For lists of detected and suppressed errors, rerun with: -s  
==48402== ERROR SUMMARY: 202427 errors from 18 contexts (suppressed: 314235 from 353)
```

Above is the error count after running DRD. There were no critical errors, as the program was able to complete itself.

Step 3: Callgrind and callgrind_annotate

After timing the program several times, the average time was determined to be about 16 minutes and 50 seconds. This was due to running 1000 times, and there was about a 1 second delay between the agent placing the items and the chef taking it. This 1 second delay, while small, can clearly cause many time delays.

This seems like a good value as it takes about 1 second from the agent placing the item to the sandwich being made, it just seems like a long time due to the sheer number of sandwiches being made.

According to the annotated callgrind, the function spends about 2% of the total time in the put and get functions, with an additional 2% sleeping, and 1% in the wait_for function. Below is the full output of the 100% of the runtime, it is fairly long.

```
1,260,309 ( 6.60%) ./libio/.libio/iofwrite.c:fwrite [/usr/lib/x86_64-linux-gnu/libc.so.6]

 987,152 ( 5.17%) ./libio/.libio/fileops.c:_IO_file_xsputn@@GLIBC_2.2.5 [/usr/lib/x86_64-linux-
gnu/libc.so.6]

 969,171 ( 5.07%) ???::std::basic_ostream<char, std::char_traits<char> >&
std::__ostream_insert<char, std::char_traits<char> >(std::basic_ostream<char,
std::char_traits<char> >&, char const*, long) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

 638,185 ( 3.34%) ???::std::ostream::sentry::sentry(std::ostream&) [/usr/lib/x86_64-linux-
gnu/libstdc++.so.6.0.33]

 503,306 ( 2.64%) ./elf/.elf/dl-lookup.c:do_lookup_x [/usr/lib/x86_64-linux-gnu/ld-linux-x86-
64.so.2]

 498,994 ( 2.61%) ./elf/./sysdeps/generic/dl-new-hash.h:_dl_lookup_symbol_x

 368,000 ( 1.93%) ???::std::ostreambuf_iterator<char, std::char_traits<char> > std::num_put<char,
std::ostreambuf_iterator<char, std::char_traits<char> >
>::_M_insert_int<long>(std::ostreambuf_iterator<char, std::char_traits<char> >, std::ios_base&,
char, long) const [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

 349,676 ( 1.83%) ./string/./sysdeps/x86_64/multiarch/memmove-vec-unaligned-
erms.S:__mempcpy_avx_unaligned_erms [/usr/lib/x86_64-linux-gnu/libc.so.6]

 339,507 ( 1.78%)
./time/./sysdeps/unix/sysv/linux/clock_nanosleep.c:clock_nanosleep@@GLIBC_2.17
[/usr/lib/x86_64-linux-gnu/libc.so.6]

 336,308 ( 1.76%) ./libio/.libio/libioP.h:fwrite

 232,021 ( 1.21%) ???::std::ostream& std::ostream::_M_insert<long>(long) [/usr/lib/x86_64-linux-
gnu/libstdc++.so.6.0.33]

 224,061 ( 1.17%) ./libio/.libio/fileops.c:_IO_do_write@@GLIBC_2.2.5 [/usr/lib/x86_64-linux-
gnu/libc.so.6]

 214,368 ( 1.12%) ???::std::chrono::duration<long, std::ratio<1l, 1000l> >::count() const
[/home/noatley/3303/Assignment1/Assignment1]

 214,368 ( 1.12%) ???::void std::this_thread::sleep_for<long, std::ratio<1l, 1000l>
>(std::chrono::duration<long, std::ratio<1l, 1000l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]
```

207,796 (1.09%) ./nptl./nptl/pthread_mutex_lock.c:pthread_mutex_lock@@GLIBC_2.2.5
[usr/lib/x86_64-linux-gnu/libc.so.6]

200,040 (1.05%) ???::std::basic_ostream<char, std::char_traits<char> >& std::operator<<
<std::char_traits<char> >(std::basic_ostream<char, std::char_traits<char> >&, char const*)
[usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

197,040 (1.03%) ???::std::chrono::duration<long, std::ratio<1l, 1l> >::count() const
[home/noatley/3303/Assignment1/Assignment1]

187,401 (0.98%) ./elf./elf/dl_lookup.c:_dl_lookup_symbol_x [usr/lib/x86_64-linux-gnu/ld-linux-
x86-64.so.2]

184,688 (0.97%) ./nptl./nptl/cancellation.c:__pthread_enable_asynccancel [usr/lib/x86_64-
linux-gnu/libc.so.6]

176,044 (0.92%) ???::std::ostream::put(char) [usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

174,872 (0.92%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::_M_get_insert_unique_pos(int const&) [home/noatley/3303/Assignment1/Assignment1]

174,040 (0.91%) ???::Agent<int>::operator()(std::__cxx11::basic_string<char,
std::char_traits<char>, std::allocator<char> > const&)
[home/noatley/3303/Assignment1/Assignment1]

173,145 (0.91%) ./nptl./nptl/cancellation.c:__pthread_disable_asynccancel [usr/lib/x86_64-
linux-gnu/libc.so.6]

164,174 (0.86%) ???::std::ostream::flush() [usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

161,375 (0.84%) ???::Chef<int>::operator()(std::__cxx11::basic_string<char,
std::char_traits<char>, std::allocator<char> > const&)
[home/noatley/3303/Assignment1/Assignment1]

161,004 (0.84%) ???::0x0000000000146900 [usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

160,040 (0.84%) ./libio./libio/putc.c:putc [usr/lib/x86_64-linux-gnu/libc.so.6]

158,424 (0.83%) ???::void std::this_thread::sleep_for<long, std::ratio<1l, 1l>
>(std::chrono::duration<long, std::ratio<1l, 1l> > const&)
[home/noatley/3303/Assignment1/Assignment1]

156,488 (0.82%) ???::int
std::uniform_int_distribution<int>::operator()(std::linear_congruential_engine<unsigned long,
16807ul, 0ul, 2147483647ul> >(std::linear_congruential_engine<unsigned long, 16807ul, 0ul,
2147483647ul>&, std::uniform_int_distribution<int>::param_type const&)
[home/noatley/3303/Assignment1/Assignment1]

153,705 (0.80%) ./nptl./nptl/pthread_mutex_unlock.c:pthread_mutex_unlock@@GLIBC_2.2.5
[usr/lib/x86_64-linux-gnu/libc.so.6]

153,120 (0.80%) ???::std::common_type<std::chrono::duration<long, std::ratio<1l, 1000l> >, std::chrono::duration<long, std::ratio<1l, 1l> >::type std::chrono::operator-<long, std::ratio<1l, 1000l>, long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&, std::chrono::duration<long, std::ratio<1l, 1l> > const&)

[/home/noatley/3303/Assignment1/Assignment1]

152,038 (0.80%) ./libio/./libio/fileops.c:_IO_file_write@@GLIBC_2.2.5 [/usr/lib/x86_64-linux-gnu/libc.so.6]

149,106 (0.78%) ./string/./sysdeps/x86_64/multiarch/strlen-avx2.S:__strlen_avx2
[/usr/lib/x86_64-linux-gnu/libc.so.6]

147,743 (0.77%) ./elf/./elf/dl-reloc.c:_dl_relocate_object [/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]

137,808 (0.72%) ???::std::chrono::duration<long, std::ratio<1l, 1000l> >::duration<long, void>(long const&) [/home/noatley/3303/Assignment1/Assignment1]

136,374 (0.71%) ./libio/./libio/ioflush.c:fflush [/usr/lib/x86_64-linux-gnu/libc.so.6]

134,532 (0.70%) ???::std::pair<std::_Rb_tree_iterator<int>, bool> std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_insert_unique<int const&>(int const&)
[/home/noatley/3303/Assignment1/Assignment1]

133,481 (0.70%) ???::std::_Rb_tree_iterator<int> std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_insert<int const&, std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_Alloc_node>(std::_Rb_tree_node_base*, std::_Rb_tree_node_base*, int const&, std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_Alloc_node&) [/home/noatley/3303/Assignment1/Assignment1]

126,018 (0.66%) ???::__gnu_cxx::stdio_sync_filebuf<char, std::char_traits<char> >::xsputn(char const*, long) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

122,496 (0.64%) ???::std::chrono::duration<long, std::ratio<1l, 1l> >
std::chrono::__duration_cast_impl<std::chrono::duration<long, std::ratio<1l, 1l> >, std::ratio<1l, 1000l>, long, true, false>::__cast<long, std::ratio<1l, 1000l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&) [/home/noatley/3303/Assignment1/Assignment1]

118,668 (0.62%) ???::bool std::chrono::operator< <long, std::ratio<1l, 1000l>, long, std::ratio<1l, 1000l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&, std::chrono::duration<long, std::ratio<1l, 1000l> > const&) [/home/noatley/3303/Assignment1/Assignment1]

114,286 (0.60%) ./elf/./elf/dl-lookup.c:check_match [/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]

113,832 (0.60%) ???::std::chrono::duration<long, std::ratio<1l, 1l> >::duration<long, void>(long const&) [/home/noatley/3303/Assignment1/Assignment1]

112,259 (0.59%) ./libio/./libio/fileops.c:_IO_file_sync@@GLIBC_2.2.5 [/usr/lib/x86_64-linux-gnu/libc.so.6]

112,153 (0.59%) ./elf/./sysdeps/x86_64/dl-machine.h:_dl_relocate_object

111,158 (0.58%) ./elf/./elf/do-rel.h:_dl_relocate_object

108,890 (0.57%) ./malloc/./malloc/malloc.c:_int_free [/usr/lib/x86_64-linux-gnu/libc.so.6]

107,502 (0.56%) ???::std::common_type<std::chrono::duration<long, std::ratio<1l, 1l> >, std::chrono::duration<long, std::ratio<1l, 1l> >::type std::chrono::operator-(<long, std::ratio<1l, 1l>, long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&, std::chrono::duration<long, std::ratio<1l, 1l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

106,000 (0.55%) ???::void std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_construct_node<int const&>(std::_Rb_tree_node<int>*, int const&)
[/home/noatley/3303/Assignment1/Assignment1]

101,545 (0.53%) ./nptl/./nptl/ptthread_cond_wait.c:pthread_cond_wait@@GLIBC_2.3.2
[/usr/lib/x86_64-linux-gnu/libc.so.6]

98,000 (0.51%) ???::int const& std::forward<int const&>(std::remove_reference<int const&>::type&) [/home/noatley/3303/Assignment1/Assignment1]

96,024 (0.50%) ./io/./sysdeps/unix/sysv/linux/write.c:write [/usr/lib/x86_64-linux-gnu/libc.so.6]

91,872 (0.48%) ???::std::chrono::duration<long, std::ratio<1l, 1000000000l> > std::chrono::__duration_cast_impl<std::chrono::duration<long, std::ratio<1l, 1000000000l> >, std::ratio<1000000l, 1l>, long, false, true>::__cast<long, std::ratio<1l, 1000l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

91,872 (0.48%) ???::std::chrono::duration<long, std::ratio<1l, 1000l> > std::chrono::__duration_cast_impl<std::chrono::duration<long, std::ratio<1l, 1000l> >, std::ratio<1000l, 1l>, long, false, true>::__cast<long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&) [/home/noatley/3303/Assignment1/Assignment1]

91,872 (0.48%) ???::std::chrono::duration<long, std::ratio<1l, 1000l> >::duration<long, std::ratio<1l, 1l>, void>(std::chrono::duration<long, std::ratio<1l, 1l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

90,402 (0.47%) ./malloc/./malloc/malloc.c:malloc [/usr/lib/x86_64-linux-gnu/libc.so.6]

88,000 (0.46%) ???::0x00000000000142070 [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

87,699 (0.46%) ???::bool std::chrono::operator< <long, std::ratio<1l, 1l>, long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&, std::chrono::duration<long, std::ratio<1l, 1l> > const&) [/home/noatley/3303/Assignment1/Assignment1]

86,490 (0.45%) ???::__pthread_mutex_lock(pthread_mutex_t*)
[/home/noatley/3303/Assignment1/Assignment1]

86,490 (0.45%) ???:_gthread_mutex_unlock(pthread_mutex_t*)
[/home/noatley/3303/Assignment1/Assignment1]

84,021 (0.44%) ???:std::basic_ostream<char, std::char_traits<char> >& std::endl<char, std::char_traits<char> >(std::basic_ostream<char, std::char_traits<char> >&) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

80,724 (0.42%) ???:std::mutex::lock() [/home/noatley/3303/Assignment1/Assignment1]

79,884 (0.42%) ./posix/./sysdeps/unix/sysv/linux/nanosleep.c:nanosleep [/usr/lib/x86_64-linux-gnu/libc.so.6]

79,884 (0.42%) ???:std::chrono::duration<long, std::ratio<1l, 1000000000l> >::duration<long, void>(long const&) [/home/noatley/3303/Assignment1/Assignment1]

76,560 (0.40%) ???:std::chrono::duration<long, std::ratio<1l, 1000l> >::zero()
[/home/noatley/3303/Assignment1/Assignment1]

75,720 (0.40%) ???:std::set<int, std::less<int>, std::allocator<int> >::insert(int const&)
[/home/noatley/3303/Assignment1/Assignment1]

74,907 (0.39%) ???:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_S_key(std::_Rb_tree_node<int> const*) [/home/noatley/3303/Assignment1/Assignment1]

71,819 (0.38%) ???:std::_Rb_tree_iterator<int>::_Rb_tree_iterator(std::_Rb_tree_node_base*)
[/home/noatley/3303/Assignment1/Assignment1]

71,615 (0.37%) ./string/./sysdeps/x86_64/multiarch/./multiarch/strcmp-sse2.S:strcmp
[/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]

69,052 (0.36%) ???:Agent<int>::chooseFoods()
[/home/noatley/3303/Assignment1/Assignment1]

68,313 (0.36%) ./libio/./libio/libioP.h:fflush

67,896 (0.36%) ???:std::chrono::duration<long, std::ratio<1l, 1000000000l> >
std::chrono::__duration_cast_impl<std::chrono::duration<long, std::ratio<1l, 1000000000l> >, std::ratio<1000000000l, 1l>, long, false, true>::__cast<long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

66,309 (0.35%) ???:std::unique_lock<std::mutex>::unlock()
[/home/noatley/3303/Assignment1/Assignment1]

65,624 (0.34%) ???:std::__detail::_Mod<unsigned long, 2147483647ul, 16807ul, 0ul, true, true>::__calc(unsigned long) [/home/noatley/3303/Assignment1/Assignment1]

64,459 (0.34%) ./nptl/./nptl/pthread_cond_common.c:pthread_cond_broadcast@@GLIBC_2.3.2

63,426 (0.33%) ???:std::mutex::unlock() [/home/noatley/3303/Assignment1/Assignment1]

63,426 (0.33%) ???::std::unique_lock<std::mutex>::lock()
[/home/noatley/3303/Assignment1/Assignment1]

63,100 (0.33%) ???::std::pair<std::_Rb_tree_const_iterator<int>, bool>::pair<std::_Rb_tree_iterator<int>&, bool&, true>(std::_Rb_tree_iterator<int>&, bool&)
[/home/noatley/3303/Assignment1/Assignment1]

61,237 (0.32%) ???::std::_Rb_tree_node<int>::_M_valptr() const
[/home/noatley/3303/Assignment1/Assignment1]

60,543 (0.32%) ???::std::unique_lock<std::mutex>::unique_lock(std::mutex&)
[/home/noatley/3303/Assignment1/Assignment1]

60,128 (0.31%)
./nptl/./nptl/cond_broadcast.c:pthread_cond_broadcast@@GLIBC_2.3.2 [/usr/lib/x86_64-linux-gnu/libc.so.6]

58,021 (0.30%) ???::0x00000000048f6360 [???]

57,660 (0.30%) ???::__gthread_active_p() [/home/noatley/3303/Assignment1/Assignment1]

57,181 (0.30%) ???::Box<int>::put(int, int) [/home/noatley/3303/Assignment1/Assignment1]

56,728 (0.30%) ???::std::_Identity<int>::operator()(int const&) const
[/home/noatley/3303/Assignment1/Assignment1]

56,580 (0.30%) ???::std::chrono::duration<long, std::ratio<1l, 1l> >::zero()
[/home/noatley/3303/Assignment1/Assignment1]

56,324 (0.29%) ./malloc/./malloc/malloc.c:free [/usr/lib/x86_64-linux-gnu/libc.so.6]

56,000 (0.29%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_get_Node_allocator() [/home/noatley/3303/Assignment1/Assignment1]

56,000 (0.29%) ???::std::_V2::condition_variable_any::notify_all()
[/home/noatley/3303/Assignment1/Assignment1]

55,670 (0.29%) ???::__gnu_cxx::__aligned_membuf<int>::_M_ptr() const
[/home/noatley/3303/Assignment1/Assignment1]

55,528 (0.29%) ???::std::pair<std::_Rb_tree_iterator<int>, bool>::pair<std::_Rb_tree_iterator<int>, bool, true>(std::_Rb_tree_iterator<int>&&, bool&&)
[/home/noatley/3303/Assignment1/Assignment1]

54,000 (0.28%) ???::std::_new_allocator<std::_Rb_tree_node<int> >::allocate(unsigned long, void const*) [/home/noatley/3303/Assignment1/Assignment1]

53,841 (0.28%) ???::std::unique_lock<std::mutex>::~unique_lock()
[/home/noatley/3303/Assignment1/Assignment1]

53,592 (0.28%) ???:bool std::chrono::operator<=<long, std::ratio<1l, 1000l>, long, std::ratio<1l, 1000l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&, std::chrono::duration<long, std::ratio<1l, 1000l> > const&) [/home/noatley/3303/Assignment1/Assignment1]

53,256 (0.28%) ???:std::chrono::duration<long, std::ratio<1l, 1000000000l> >::count() const
[/home/noatley/3303/Assignment1/Assignment1]

52,980 (0.28%) ???:void std::_V2::condition_variable_any::wait<std::mutex>(std::mutex&)
[/home/noatley/3303/Assignment1/Assignment1]

52,000 (0.27%) ???:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::_M_destroy_node(std::_Rb_tree_node<int>*) [/home/noatley/3303/Assignment1/Assignment1]

49,938 (0.26%) ???:std::less<int>::operator()(int const&, int const&) const
[/home/noatley/3303/Assignment1/Assignment1]

49,764 (0.26%) ???:std::chrono::duration<long, std::ratio<1l, 1000l> >::duration<int, void>(int
const&) [/home/noatley/3303/Assignment1/Assignment1]

48,093 (0.25%) ???:Box<int>::boxInquiry(int&, int&)
[/home/noatley/3303/Assignment1/Assignment1]

48,012 (0.25%) ./libio/.libio/libioP.h:putc

48,000 (0.25%) ???:void std::__advance<std::_Rb_tree_const_iterator<int>,
long>(std::_Rb_tree_const_iterator<int>&, long, std::bidirectional_iterator_tag)
[/home/noatley/3303/Assignment1/Assignment1]

46,000 (0.24%) ???:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::_M_put_node(std::_Rb_tree_node<int>*) [/home/noatley/3303/Assignment1/Assignment1]

45,361 (0.24%) ???:Chef<int>::checkFood(int, int)
[/home/noatley/3303/Assignment1/Assignment1]

44,250 (0.23%) ./nptl/.nptl/futex-internal.c: __futex_abstimed_wait_cancelable64
[/usr/lib/x86_64-linux-gnu/libc.so.6]

44,101 (0.23%) ???:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::_M_erase(std::_Rb_tree_node<int>*) [/home/noatley/3303/Assignment1/Assignment1]

44,000 (0.23%) ???:std::_Rb_tree_node<int>* std::_Rb_tree<int, int, std::_Identity<int>,
std::less<int>, std::allocator<int> >::_M_create_node<int const&>(int const&)
[/home/noatley/3303/Assignment1/Assignment1]

44,000 (0.23%) ???:std::_Rb_tree_node<int>::_M_valptr()
[/home/noatley/3303/Assignment1/Assignment1]

44,000 (0.23%) ???:std::ostream::operator<<(int) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

44,000 (0.23%) ???:std::pair<std::_Rb_tree_node_base*,
std::_Rb_tree_node_base*>::pair<std::_Rb_tree_node<int>* &, std::_Rb_tree_node_base*&,>


```

true>(std::_Rb_tree_node<int>*&, std::_Rb_tree_node_base*&)
[/home/noatley/3303/Assignment1/Assignment1]

43,245 ( 0.23%) ???:_shared_ptr_access<std::mutex, (__gnu_cxx::Lock_policy)2, false,
false>::operator*() const [/home/noatley/3303/Assignment1/Assignment1]

43,000 ( 0.23%) ???::Box<int>::get() [/home/noatley/3303/Assignment1/Assignment1]

42,011 ( 0.22%) ???::0x00000000048f5d00 [???]

42,004 ( 0.22%) ???::0x0000000004b25430 [???]

40,663 ( 0.21%) ???::std::_Sp_counted_base<(__gnu_cxx::Lock_policy)2>::_M_release()
[/home/noatley/3303/Assignment1/Assignment1]

40,384 ( 0.21%) ???::std::uniform_int_distribution<int>::param_type::a() const
[/home/noatley/3303/Assignment1/Assignment1]

40,000 ( 0.21%) ???::__gnu_cxx::__aligned_membuf<int>::_M_ptr()
[/home/noatley/3303/Assignment1/Assignment1]

40,000 ( 0.21%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::begin() const [/home/noatley/3303/Assignment1/Assignment1]

39,962 ( 0.21%) ???::std::_Rb_tree_insert_and_rebalance(bool, std::_Rb_tree_node_base*,
std::_Rb_tree_node_base*, std::_Rb_tree_node_base&) [/usr/lib/x86_64-linux-
gnu/libstdc++.so.6.0.33]

39,942 ( 0.21%) ???::std::chrono::duration_values<long>::zero()
[/home/noatley/3303/Assignment1/Assignment1]

39,606 ( 0.21%) ???::bool std::chrono::operator<=<long, std::ratio<1l, 1l>, long, std::ratio<1l, 1l>
>(std::chrono::duration<long, std::ratio<1l, 1l> > const&, std::chrono::duration<long, std::ratio<1l,
1l> > const&) [/home/noatley/3303/Assignment1/Assignment1]

38,969 ( 0.20%) ???::__gnu_cxx::__aligned_membuf<int>::_M_addr() const
[/home/noatley/3303/Assignment1/Assignment1]

38,280 ( 0.20%) ???::std::enable_if<std::chrono::__is_duration<std::chrono::duration<long,
std::ratio<1l, 1000000000l> > >::value, std::chrono::duration<long, std::ratio<1l, 1000000000l> >
>::type std::chrono::duration_cast<std::chrono::duration<long, std::ratio<1l, 1000000000l> >, long,
std::ratio<1l, 1000l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

38,280 ( 0.20%) ???::std::enable_if<std::chrono::__is_duration<std::chrono::duration<long,
std::ratio<1l, 1000l> > >::value, std::chrono::duration<long, std::ratio<1l, 1000l> > >::type
std::chrono::duration_cast<std::chrono::duration<long, std::ratio<1l, 1000l> >, long, std::ratio<1l,
1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

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38,280 (0.20%) ???::std::enable_if<std::chrono::__is_duration<std::chrono::duration<long, std::ratio<1l, 1l> >::value, std::chrono::duration<long, std::ratio<1l, 1l> >::type std::chrono::duration_cast<std::chrono::duration<long, std::ratio<1l, 1l> >, long, std::ratio<1l, 1000l> >(std::chrono::duration<long, std::ratio<1l, 1000l> > const&)
[/home/noatley/3303/Assignment1/Assignment1]

38,000 (0.20%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_drop_node(std::_Rb_tree_node<int>*) [/home/noatley/3303/Assignment1/Assignment1]

38,000 (0.20%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_get_node() [/home/noatley/3303/Assignment1/Assignment1]

38,000 (0.20%) ???::std::_new_allocator<std::_Rb_tree_node<int> >::deallocate(std::_Rb_tree_node<int>*, unsigned long)
[/home/noatley/3303/Assignment1/Assignment1]

37,860 (0.20%) ???::std::linear_congruential_engine<unsigned long, 16807ul, 0ul, 2147483647ul>::operator>() [/home/noatley/3303/Assignment1/Assignment1]

36,899 (0.19%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_erase(std::_Rb_tree_node<int>*)'2 [/home/noatley/3303/Assignment1/Assignment1]

36,777 (0.19%) ???::std::chrono::duration<long, std::ratio<1l, 1l> >::duration<int, void>(int const&) [/home/noatley/3303/Assignment1/Assignment1]

36,192 (0.19%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_end() [/home/noatley/3303/Assignment1/Assignment1]

36,000 (0.19%) ???::std::_Rb_tree_node<int>* std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_Alloc_node::operator()<int const&>(int const&) const
[/home/noatley/3303/Assignment1/Assignment1]

35,336 (0.19%) ???::int
std::uniform_int_distribution<int>::operator()<std::linear_congruential_engine<unsigned long, 16807ul, 0ul, 2147483647ul> >(std::linear_congruential_engine<unsigned long, 16807ul, 0ul, 2147483647ul>&) [/home/noatley/3303/Assignment1/Assignment1]

35,240 (0.18%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_M_begin() [/home/noatley/3303/Assignment1/Assignment1]

34,011 (0.18%) ???::0x00000000048f4b20 [???]

32,080 (0.17%) ???::operator new(unsigned long) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

32,016 (0.17%) ???::operator new(unsigned long, void*)
[/home/noatley/3303/Assignment1/Assignment1]

32,000 (0.17%) ???::std::lock_guard<std::mutex>::lock_guard(std::mutex&)
[/home/noatley/3303/Assignment1/Assignment1]

30,288 (0.16%)
 ???::std::_Rb_tree_const_iterator<int>::_Rb_tree_const_iterator(std::_Rb_tree_iterator<int> const&)
 [/home/noatley/3303/Assignment1/Assignment1]

29,708 (0.16%) ./elf/./elf/dl-tunables.c: __GI__tunables_init [/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]

29,620 (0.16%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>>::begin() [/home/noatley/3303/Assignment1/Assignment1]

28,830 (0.15%) ???::std::__shared_ptr_access<std::mutex, (__gnu_cxx::__Lock_policy)2, false, false>::_M_get() const [/home/noatley/3303/Assignment1/Assignment1]

28,290 (0.15%) ???::std::enable_if<std::chrono::__is_duration<std::chrono::duration<long, std::ratio<1l, 10000000000l> >::value, std::chrono::duration<long, std::ratio<1l, 10000000000l> >::type std::chrono::duration_cast<std::chrono::duration<long, std::ratio<1l, 10000000000l> >, long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&)
 [/home/noatley/3303/Assignment1/Assignment1]

28,192 (0.15%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>>::_M_mbegin() const [/home/noatley/3303/Assignment1/Assignment1]

28,007 (0.15%) ???::__gnu_cxx::stdio_sync_filebuf<char, std::char_traits<char> >::overflow(int)
 [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

28,004 (0.15%) ???::std::basic_ostream<char, std::char_traits<char> >& std::operator<< <char, std::char_traits<char>, std::allocator<char> >(std::basic_ostream<char, std::char_traits<char> >&, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> > const&)
 [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

28,000 (0.15%) ???::__gnu_cxx::__aligned_membuf<int>::_M_addr()
 [/home/noatley/3303/Assignment1/Assignment1]

27,764 (0.15%) ???::unsigned long std::__detail::__mod<unsigned long, 2147483647ul, 16807ul, 0ul>(unsigned long) [/home/noatley/3303/Assignment1/Assignment1]

26,598 (0.14%) ./nptl/./nptl/pthread_mutex_lock.c: __pthread_mutex_cond_lock
 [/usr/lib/x86_64-linux-gnu/libc.so.6]

24,948 (0.13%) ???::std::locale::id::_M_id() const [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

24,344 (0.13%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>>::_S_right(std::_Rb_tree_node_base*) [/home/noatley/3303/Assignment1/Assignment1]

24,000 (0.13%) ???::std::lock_guard<std::mutex>::~~lock_guard()
 [/home/noatley/3303/Assignment1/Assignment1]

23,841 (0.12%) ./nptl/./nptl/pthread_mutex_unlock.c: __pthread_mutex_unlock_usercnt
 [/usr/lib/x86_64-linux-gnu/libc.so.6]

23,346 (0.12%) ???std::__is_constant_evaluated()
[home/noatley/3303/Assignment1/Assignment1]

23,086 (0.12%) ./nptl/./nptl/descr.h: __pthread_enable_asynccancel

23,064 (0.12%) ???std::__shared_ptr<std::mutex, (__gnu_cxx::__Lock_policy)2>::get() const
[home/noatley/3303/Assignment1/Assignment1]

22,632 (0.12%) ???std::enable_if<std::chrono::__is_duration<std::chrono::duration<long, std::ratio<1l, 1l> >::value, std::chrono::duration<long, std::ratio<1l, 1l> >::type std::chrono::duration_cast<std::chrono::duration<long, std::ratio<1l, 1l> >, long, std::ratio<1l, 1l> >(std::chrono::duration<long, std::ratio<1l, 1l> > const&)
[home/noatley/3303/Assignment1/Assignment1]

22,075 (0.12%) ???std::_Sp_counted_base<(__gnu_cxx::__Lock_policy)2>::_M_add_ref_copy()
[home/noatley/3303/Assignment1/Assignment1]

22,059 (0.12%) ./malloc/./malloc/arena.c:free

22,000 (0.12%) ???std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_Alloc_node::_Alloc_node(std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >&) [home/noatley/3303/Assignment1/Assignment1]

22,000 (0.12%)
???std::_Rb_tree_const_iterator<int>::_Rb_tree_const_iterator(std::_Rb_tree_node_base const*)
[home/noatley/3303/Assignment1/Assignment1]

22,000 (0.12%) ???std::_Rb_tree_const_iterator<int>::operator*() const
[home/noatley/3303/Assignment1/Assignment1]

21,336 (0.11%) ???bool&& std::forward<bool>(std::remove_reference<bool>::type&)
[home/noatley/3303/Assignment1/Assignment1]

21,000 (0.11%) ???std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_Rb_tree_impl<std::less<int>, true>::_Rb_tree_impl()
[home/noatley/3303/Assignment1/Assignment1]

21,000 (0.11%) ???std::uniform_int_distribution<int>::param_type::param_type(int, int)
[home/noatley/3303/Assignment1/Assignment1]

20,430 (0.11%) ???std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int> >::_S_key(std::_Rb_tree_node_base const*) [home/noatley/3303/Assignment1/Assignment1]

20,192 (0.11%) ???std::uniform_int_distribution<int>::param_type::b() const
[home/noatley/3303/Assignment1/Assignment1]

20,181 (0.11%) ???std::mutex* std::__addressof<std::mutex>(std::mutex&)
[home/noatley/3303/Assignment1/Assignment1]

20,009 (0.10%) ???0x0000000048f3cf0 [??]

20,005 (0.10%) ./libio/./libio/libioP.h:_IO_do_write@@GLIBC_2.2.5

20,004 (0.10%) ????:0x0000000000010a4a0 [???

20,000 (0.10%) ????:std::set<int, std::less<int>, std::allocator<int> >::begin() const
[/home/noatley/3303/Assignment1/Assignment1]

19,848 (0.10%) ????:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::_S_left(std::_Rb_tree_node_base*) [/home/noatley/3303/Assignment1/Assignment1]

19,253 (0.10%) ????:std::operator==(std::_Rb_tree_iterator<int> const&,
std::_Rb_tree_iterator<int> const&) [/home/noatley/3303/Assignment1/Assignment1]

19,000 (0.10%) ????:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>
>::~~_Rb_tree() [/home/noatley/3303/Assignment1/Assignment1]

18,543 (0.10%)
????:std::__shared_count<(__gnu_cxx::Lock_policy)2>::__shared_count(std::__shared_count<(__g
nu_cxx::Lock_policy)2> const&) [/home/noatley/3303/Assignment1/Assignment1]

17,668 (0.09%) ????:bool& std::forward<bool&>(std::remove_reference<bool&>::type&)
[/home/noatley/3303/Assignment1/Assignment1]

17,668 (0.09%) ????:std::_Rb_tree_iterator<int>&
std::forward<std::_Rb_tree_iterator<int>&>(std::remove_reference<std::_Rb_tree_iterator<int>&>::t
ype&) [/home/noatley/3303/Assignment1/Assignment1]

17,668 (0.09%) ????:std::_Rb_tree_iterator<int>&& std::forward<std::_Rb_tree_iterator<int>
>(std::remove_reference<std::_Rb_tree_iterator<int> >::type&)
[/home/noatley/3303/Assignment1/Assignment1]

17,660 (0.09%) ????:std::__shared_ptr<std::mutex,
(__gnu_cxx::Lock_policy)2>::__shared_ptr(std::__shared_ptr<std::mutex,
(__gnu_cxx::Lock_policy)2> const&) [/home/noatley/3303/Assignment1/Assignment1]

17,660 (0.09%)
????:std::unique_lock<std::mutex>::unique_lock(std::unique_lock<std::mutex>&&)
[/home/noatley/3303/Assignment1/Assignment1]

17,000 (0.09%) ????:std::_Rb_tree_header::_M_reset()
[/home/noatley/3303/Assignment1/Assignment1]

16,648 (0.09%) ./elf/./elf/dl-version.c:_dl_check_map_versions [/usr/lib/x86_64-linux-gnu/ld-
linux-x86-64.so.2]

16,000 (0.08%) ????:std::uniform_int_distribution<int>::uniform_int_distribution(int, int)
[/home/noatley/3303/Assignment1/Assignment1]

15,011 (0.08%) ????:std::_V2::condition_variable_any::_Unlock<std::mutex>::~~_Unlock()
[/home/noatley/3303/Assignment1/Assignment1]

14,144 (0.07%) ???:std::__shared_count<(__gnu_cxx::Lock_policy)2>::~__shared_count()
[/home/noatley/3303/Assignment1/Assignment1]

14,002 (0.07%) ???:0x0000000000010a480 [???]

14,000 (0.07%) ???:std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>>::_Rb_tree_impl<std::less<int>, true>::~~_Rb_tree_impl()
[/home/noatley/3303/Assignment1/Assignment1]

14,000 (0.07%) ???:std::_Rb_tree_const_iterator<int>::operator++()
[/home/noatley/3303/Assignment1/Assignment1]

14,000 (0.07%) ???:std::_Rb_tree_node<int>* &
std::forward<std::_Rb_tree_node<int>* &>(std::remove_reference<std::_Rb_tree_node<int>* &>::type &)
[/home/noatley/3303/Assignment1/Assignment1]

14,000 (0.07%) ???:std::_Rb_tree_node_base* &
std::forward<std::_Rb_tree_node_base* &>(std::remove_reference<std::_Rb_tree_node_base* &>::type &)
[/home/noatley/3303/Assignment1/Assignment1]

13,314 (0.07%) ???:0x0000000000010a380 [???]

13,245 (0.07%)
??? :std::_V2::condition_variable_any::_Unlock<std::mutex>::_Unlock(std::mutex &)
[/home/noatley/3303/Assignment1/Assignment1]

13,000 (0.07%) ???:std::_Rb_tree_header::_Rb_tree_header()
[/home/noatley/3303/Assignment1/Assignment1]

12,362 (0.06%) ???:std::shared_ptr<std::mutex>::shared_ptr(std::shared_ptr<std::mutex> const &)
[/home/noatley/3303/Assignment1/Assignment1]

12,192 (0.06%) ???:std::locale::_Impl::_M_install_facet(std::locale::id const*, std::locale::facet const*)
[/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

12,018 (0.06%) ???:__gnu_cxx::stdio_sync_filebuf<char, std::char_traits<char> >::sync()
[/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

12,000 (0.06%) ???:std::num_put<char, std::ostreambuf_iterator<char, std::char_traits<char> >>::do_put(std::ostreambuf_iterator<char, std::char_traits<char> >, std::ios_base &, char, long) const
[/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

11,532 (0.06%) ???:0x0000000000010a430 [???]

11,532 (0.06%) ???:0x0000000000010a590 [???]

11,528 (0.06%) ???:std::pair<std::_Rb_tree_const_iterator<int>, bool>::operator=(std::pair<std::_Rb_tree_const_iterator<int>, bool> &&)
[/home/noatley/3303/Assignment1/Assignment1]

11,000 (0.06%) ???::std::_Rb_tree<int, int, std::_Identity<int>, std::less<int>, std::allocator<int>>::_Rb_tree() [/home/noatley/3303/Assignment1/Assignment1]

11,000 (0.06%) ???::std::set<int, std::less<int>, std::allocator<int>>::set()
[/home/noatley/3303/Assignment1/Assignment1]

11,000 (0.06%) ???::std::set<int, std::less<int>, std::allocator<int>>::~set()
[/home/noatley/3303/Assignment1/Assignment1]

10,620 (0.06%) ./elf/./sysdeps/x86_64/tls_get_addr.S:__tls_get_addr [/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]

10,608 (0.06%) ???::std::__shared_ptr<std::mutex, (__gnu_cxx::_Lock_policy)2>::~__shared_ptr()
[/home/noatley/3303/Assignment1/Assignment1]

10,596 (0.06%) ./nptl/./sysdeps/nptl/futex-internal.h:pthread_cond_broadcast@@GLIBC_2.3.2

10,487 (0.05%) ./elf/./sysdeps/generic/ldsodefs.h:_dl_relocate_object

10,038 (0.05%) ???::std::_Rb_tree_increment(std::_Rb_tree_node_base const*) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

9,724 (0.05%) ???::std::shared_ptr<std::mutex>::~shared_ptr()
[/home/noatley/3303/Assignment1/Assignment1]

9,585 (0.05%) ./elf/./sysdeps/generic/ldsodefs.h:do_lookup_x

9,216 (0.05%) ./wcsmbcs/./wcsmbcs/btowc.c:btowc [/usr/lib/x86_64-linux-gnu/libc.so.6]

8,908 (0.05%) ???::std::pair<std::_Rb_tree_node_base*, std::_Rb_tree_node_base*>::pair<std::_Rb_tree_node_base*, std::_Rb_tree_node_base*, true>(std::_Rb_tree_node_base* const&, std::_Rb_tree_node_base* const&)
[/home/noatley/3303/Assignment1/Assignment1]

8,177 (0.04%) ???:0x00000000048f3800 [???]

8,169 (0.04%) ./elf/./elf/dl-runtime.c:_dl_fixup [/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]

8,027 (0.04%) ???:0x00000000048f64b0 [???]

8,013 (0.04%) ???:0x00000000048f48a0 [???]

8,007 (0.04%) ???:0x00000000048f30f0 [???]

8,007 (0.04%) ???:0x00000000048f4b60 [???]

8,005 (0.04%) ???:0x00000000048f34c0 [???]

8,005 (0.04%) ???:0x00000000048f4c20 [???]

8,002 (0.04%) ???:0x000000000010a4e0 [???]

8,002 (0.04%) ???::std::ostream::operator<<(std::ostream& (*)(std::ostream&)) [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]

8,000 (0.04%) ???:0x0000000000010a5d0 [???]
7,965 (0.04%) ???:std::uncaught_exception() [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]
7,668 (0.04%) ./elf/./sysdeps/generic/dl-protected.h:do_lookup_x
7,080 (0.04%) ???:__cxa_get_globals [/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33]
7,000 (0.04%) ???:std::_Rb_tree_key_compare<std::less<int> >::_Rb_tree_key_compare()
[/home/noatley/3303/Assignment1/Assignment1]
7,000 (0.04%) ???:std::__new_allocator<std::_Rb_tree_node<int> >::~__new_allocator()
[/home/noatley/3303/Assignment1/Assignment1]
6,307 (0.03%) ./nptl/./nptl/lowlevellock.c:__lll_lock_wake [/usr/lib/x86_64-linux-gnu/libc.so.6]
6,216 (0.03%) ./nptl/./nptl/cleanup_compat.c:pthread_cleanup_push@@GLIBC_2.34
[/usr/lib/x86_64-linux-gnu/libc.so.6]
6,181 (0.03%) ???:std::remove_reference<std::unique_lock<std::mutex>&>::type&&
std::move<std::unique_lock<std::mutex>&>(std::unique_lock<std::mutex>&)
[/home/noatley/3303/Assignment1/Assignment1]